

Lopez Spectral Stability Theorem

Formal Statement and Proof

Joshua Lopez

DCGP Research | Distributed Constitutional Governance Protocol

U.S. Patent Application No. 19/555,951 (Utility Nonprovisional, filed March 4, 2026)

Provisional Cross-References: USPTO Nos. 63/995,404 (AURAL-M) | 63/985,742 (AURAL)

This work is licensed under CC BY-NC-ND 4.0 International.

<https://creativecommons.org/licenses/by-nc-nd/4.0/>

From: AURAL-M: A Unified Theory of Constraint-Invariant Governance for Recursive Systems (Section 10 — AURAL-M)

1. Assumptions

Let the system dynamics satisfy $\dot{x} = f(x) + d(x,t)$ with:

- (A1) $f(x)$ locally Lipschitz on compact domain D .
- (A2) Jacobian spectrum of f bounded on D .
- (A3) Disturbances bounded: $\|d(x,t)\| \leq \Delta$ for all $t \geq 0$.
- (A4) Regulator updates satisfy the Step-In-Time constraint: $|\delta_K| \leq L \cdot dt$.
- (A5) Spectral estimate obtained through renormalized perturbation evolution.

2. Regulator Structure

Spectral deviation: $S(t) = \hat{\gamma}(t) - \rho^*$. Fast authority loop:

$$\begin{aligned} \dot{K} &= \alpha \cdot S - \beta \cdot K \\ \dot{g} &= -k_g \cdot K \quad \dot{\lambda} = k_\lambda \cdot K \end{aligned}$$

Slow consequence memory:

$$\begin{aligned} \dot{M} &= \epsilon_s \cdot S - \beta_s \cdot M \\ \rho_{\text{eff}} &= \rho^* - \kappa_s \cdot M \end{aligned}$$

3. Stability Margin Definition

Define the composite stability margin:

$$\begin{aligned}\sigma(t) &= \min(\text{spectral_gap}(t), \text{lyapunov_margin}(t), \\ &\quad \text{distance_margin}(t)) \\ \delta_{\min} &= \kappa_s * \inf_{t \geq 0} \sigma(t)\end{aligned}$$

4. Theorem Statement

Theorem (Lopez Spectral Stability Theorem)

Under assumptions (A1)-(A5), there exists a positive margin $\delta_{\min} > 0$ such that if $\Delta < \delta_{\min}$, then the closed-loop system under the AURAL-M regulator satisfies:

- (1) Ultimate boundedness of state: $\limsup_{t \rightarrow \infty} \|x(t)\| \leq C$ for some finite constant C .
- (2) Spectral convergence: $\gamma_{\hat{t}} \rightarrow \rho_{\text{eff}}$ as $t \rightarrow \infty$.
- (3) Authority boundedness: $\|K(t)\| \leq K_{\max} < \infty$ for all $t \geq 0$.

5. Proof Sketch

Step 1 establishes that the renormalized FTLE estimator under (A5) produces a constitutionally valid spectral measurement — one that does not accumulate bias from the dominant tangent direction over long trajectories.

Step 2 applies a small-gain argument: the Step-in-Time constraint (A4) bounds the rate of authority escalation, preventing finite-time escape from D under the disturbance bound (A3).

Step 3 constructs the composite Lyapunov function $V(x, K, M)$ with:

$$\dot{V} \leq -c * V + \Delta^2$$

which under $\Delta < \delta_{\min}$ establishes that V remains bounded. Ultimate boundedness follows from the compactness of the level sets of V . Spectral convergence and authority boundedness follow from the LaSalle analysis. QED

6. Boundedness Type

The result is ultimate boundedness, also called input-to-state practical stability. The system may move freely during transients but eventually remains within a compact set. Asymptotic stability to a fixed point is not claimed and would not be correct under persistent bounded disturbance. The appropriate claim under persistent disturbance is ultimate boundedness — and this is what the theorem provides.

U.S. Patent Application No. 19/555,951 | Provisionals: 63/995,404 and 63/985,742
Copyright 2026 Joshua Lopez. Licensed under CC BY-NC-ND 4.0 International.
<https://creativecommons.org/licenses/by-nc-nd/4.0/>